

Multiple Choice (10 marks)

1. High entropy means that the partitions in classification are
 - (a) pure
 - (b) not pure
 - (c) useful
 - (d) useless
2. Which of the following statements about Naive Bayes is incorrect?
 - (a) Attributes are equally important.
 - (b) Attributes are statistically dependent of one another given the class value.
 - (c) Attributes are statistically independent of one another given the class value.
 - (d) Attributes can be nominal or numeric
3. Statement 1| RoBERTa pretrains on a corpus that is approximate 10 x larger than the corpus BERT pretrained on. Statement 2| ResNeXts in 2018 usually used tanh activation functions.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Which of the following is the joint probability of H, U, P, and W described by the given Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$? [note: as the product of the conditional probabilities]
 - (a) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$
 - (b) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$
 - (c) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$
 - (d) None of the above
5. Which of the following best describes what discriminative approaches try to model? (w are the parameters in the model)
 - (a) $p(y|x, w)$
 - (b) $p(y, x)$
 - (c) $p(w|x, w)$
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: Batch Normalization is used in the original ResNet paper, while Layer Normalization is not.
7. True or False: DenseNets typically require more memory than ResNets due to their dense connectivity pattern.
8. True or False: A neural network's convergence is indeed influenced by the learning rate used during training.
9. True or False: The cost of one gradient descent update, given the gradient vector g , is $O(D)$, where D is the number of dimensions.
10. True or False: Regression can be effectively used to identify direct causal relationships between input and output variables.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract all the adjacent coordinates of the given coordinate tuple.
12. Write a function to find number of odd elements in the given list using lambda function.

End of Paper